

SELF-SUPERVISED PRETRAINING FOR PROSTATE SEGMENTATION ON MICRO-ULTRASOUND IMAGES

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ABSTRACT

Expert annotation of medical images is expensive and scarce, particularly in micro-ultrasound (micro-US) guided prostate biopsy where accurate prostate segmentation requires significant urologist time. This paper investigates whether self-supervised pretraining on unlabeled micro-US scans can reduce the number of expert annotations needed to reach competitive segmentation performance. A convolutional autoencoder is pretrained on 2,152 unlabeled 2D slices from 55 patients, and the learned encoder is then attached to a U-Net decoder and fine-tuned on varying amounts of labeled data. A label-efficiency experiment at 5, 10, 20, and 45 labeled patients shows that pretraining provides a small Dice advantage at 5 and 10 patients (+0.17% and +0.28%) but this gap closes by 20 patients and reverses at 45, where the scratch baseline achieves Dice 0.862 versus 0.853 for the pretrained model. Training curves reveal that the primary benefit of pretraining is faster convergence rather than improved final accuracy. These results suggest that reconstruction-based self-supervised pretraining offers limited label-efficiency gains on small medical imaging datasets.

Index Terms— self-supervised learning, prostate segmentation, micro-ultrasound, U-Net, convolutional autoencoder

1. INTRODUCTION

Micro-ultrasound (micro-US) operates at 29 MHz, roughly four times the frequency of conventional ultrasound, yielding higher spatial resolution that enables visualization of tissue microstructure during prostate biopsy procedures. Accurate segmentation of the prostate gland on these images is clinically important for biopsy guidance, treatment planning, and volume estimation. However, obtaining expert annotations requires significant urologist time, making labeled micro-US data scarce.

Self-supervised learning (SSL) offers a potential solution by learning useful representations from unlabeled data. The idea is that an encoder pretrained to reconstruct micro-US images will learn features relevant to prostate anatomy, which can then transfer to the segmentation task when only a small number of labeled examples are available.

This work trains a convolutional autoencoder on unlabeled 2D axial slices extracted from micro-US volumes, then transfers the encoder weights into a U-Net for supervised segmentation fine-tuning. The core experiment compares the

pretrained U-Net against an identical architecture trained from scratch, varying the number of labeled patients (5, 10, 20, 45) to construct a label-efficiency curve. The hypothesis is that SSL pretraining provides the largest advantage when labeled data is limited.

2. RELATED WORK

Jiang et al. [1] proposed MicroSegNet, a supervised segmentation model trained on the same micro-US prostate dataset used in this work. Their approach uses a multi-scale attention U-Net architecture and incorporates inter-annotator disagreement to weight difficult regions during training. MicroSegNet serves as the primary supervised benchmark for this dataset.

Imran et al. [2] demonstrated that a self-supervised convolutional autoencoder can extract useful features from 2D micro-US slices for downstream prostate cancer detection tasks. Their work directly motivates our pretraining approach: rather than applying the learned features to cancer classification, we transfer them into a segmentation decoder to reduce label requirements.

Our approach differs from these works in combining the self-supervised pretraining strategy of [2] with the segmentation task and evaluation framework of [1], specifically targeting the question of how pretraining affects label efficiency on this dataset.

3. METHODOLOGY

3.1. Data pipeline

The dataset consists of 75 micro-US volumes in NIfTI format from the Micro-Ultrasound Prostate Segmentation Dataset [3], split into 55 training and 20 test patients. Each volume has spatial dimensions of 1372 x 962 pixels with a variable number of axial slices (33--45 per patient).

Each input is constructed as a 3-channel image by stacking the previous, current, and next axial slice, providing spatial context along the depth axis. All slices are resized to 256 x 256 and normalized to [0, 1]. Binary masks are resized with nearest-neighbor interpolation to preserve sharp boundaries. Fig. 1 shows a sample slice with its corresponding expert mask.

Input: 3-channel adjacent-slice stacking, resized to 256 x 256

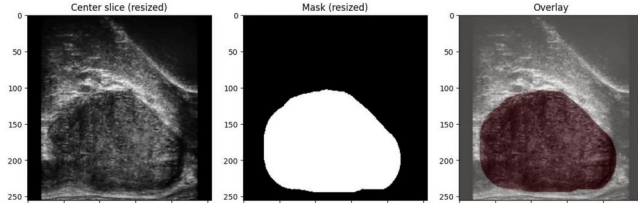


Fig. 1. Sample micro-US slice (left), expert annotation mask (center), and overlay (right).

3.2. Patient-level split

The 55 training patients are divided into three groups: all 55 are used for SSL pretraining (images only, no masks), 45 for fine-tuning with labels, and 10 for validation. The 20 test patients are held out from all training. This yields 2,152 SSL pretraining slices, 1,757 fine-tuning train slices, 395 validation slices, and 758 test slices. Splitting at the patient level prevents data leakage, since slices from the same volume are highly correlated.

3.3. Encoder architecture

The encoder has four downsampling stages. Each stage consists of two 3x3 convolution layers with batch normalization and ReLU activation, followed by 2x2 max pooling. Channel widths are 64, 128, 256, and 512 across the four stages, producing a 32 x 32 x 512 bottleneck from a 256 x 256 x 3 input. The encoder returns feature maps at each resolution level (e1 through e4) so skip connections can be used during fine-tuning.

3.4. Autoencoder pretraining

For pretraining, the encoder is paired with a simple upsampling decoder that mirrors its structure using transposed convolutions. This decoder does not use skip connections, which forces the 32 x 32 x 512 bottleneck to encode all the information needed for reconstruction. With skip connections, raw pixel information could pass directly from encoder to decoder, bypassing the bottleneck and producing a shallow representation that does not transfer well. This design follows Imran et al. [2], who used a similar architecture for feature extraction from micro-US images.

The autoencoder is trained for 50 epochs on all 2,152 SSL slices using MSE reconstruction loss, Adam optimizer with learning rate 1e-3, batch size 16, and cosine annealing learning rate schedule. Only the encoder weights are kept after pretraining.

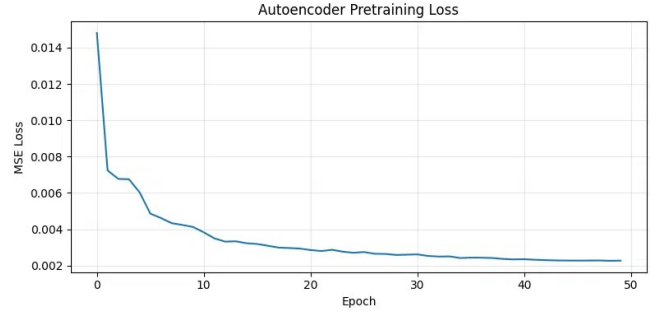


Fig. 2. Autoencoder pretraining loss over 50 epochs.

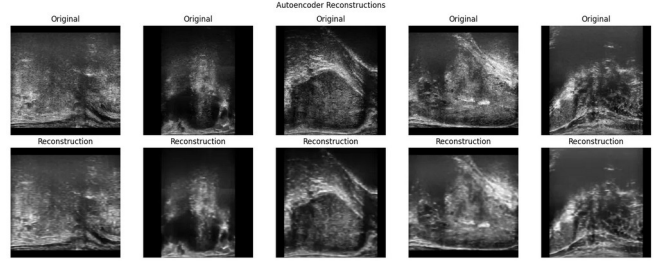


Fig. 3. Autoencoder reconstruction examples. Top: original slices. Bottom: reconstructions.

3.5. U-Net fine-tuning

For segmentation, the pretrained encoder is attached to a U-Net decoder that uses skip connections from all four encoder levels. Each decoder stage concatenates the upsampled features with the corresponding encoder features, followed by two 3x3 convolution blocks. The final layer is a 1x1 convolution producing a single-channel output.

The combined loss is Dice loss plus binary cross-entropy, following [1]. Training uses Adam optimizer with learning rate 1e-4 (lower than pretraining to preserve learned features), batch size 16, cosine annealing schedule, and runs for 100 epochs with the best model selected by validation Dice score.

4. EXPERIMENTAL RESULTS

4.1. Autoencoder pretraining

The autoencoder converged to a final MSE of 0.0023 after 50 epochs, corresponding to approximately 5% average pixel intensity error. The loss curve (Fig. 2) shows the largest drop in the first 10 epochs with diminishing returns after epoch 25. Reconstructions (Fig. 3) are visually faithful to the original slices, preserving tissue structure and prostate boundaries while slightly smoothing fine speckle detail.

4.2. Full-label segmentation (45 patients)

Both models were trained on the full 45-patient labeled set and evaluated on the 20-patient test set. The scratch U-Net slightly outperforms the pretrained model on both Dice (0.862 vs. 0.853) and HD95 (7.36 vs. 7.75 pixels). Qualitative

results (Fig. 4) show both models produce similar segmentations on mid-volume slices, with the main differences appearing on edge slices where the prostate occupies a small fraction of the image.

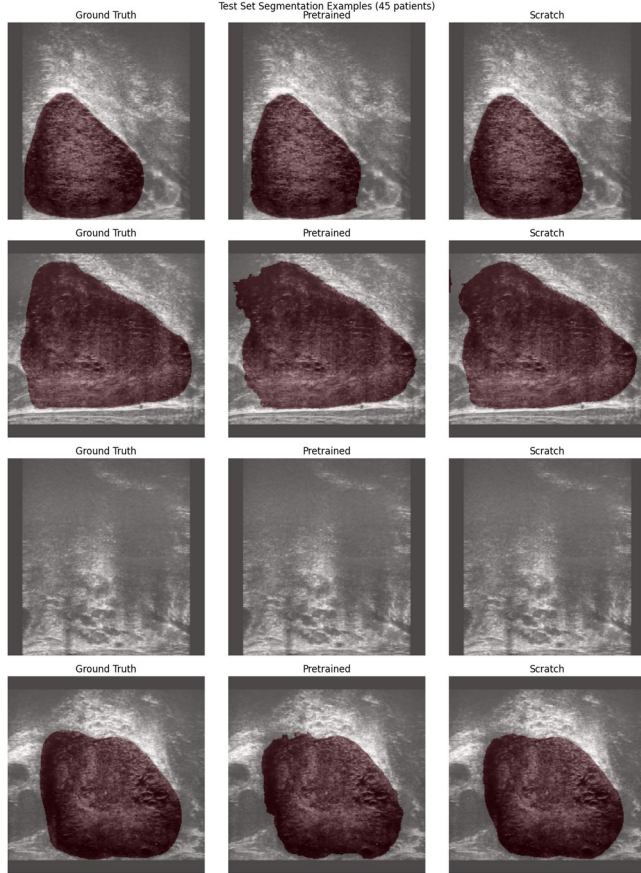


Fig. 4. Qualitative segmentation results on test slices. Columns: ground truth, pretrained prediction, scratch prediction.

4.3. Label-efficiency experiment

Both models were trained at 5, 10, 20, and 45 labeled patients and evaluated on the test set. The label-efficiency curves (Fig. 5) show that on Dice, pretrained has a small advantage at 5 patients (+0.0017) and 10 patients (+0.0028), but scratch matches at 20 and surpasses at 45. On HD95, scratch actually outperforms pretrained at 5 and 10 patients, indicating tighter boundary predictions despite slightly lower overlap.

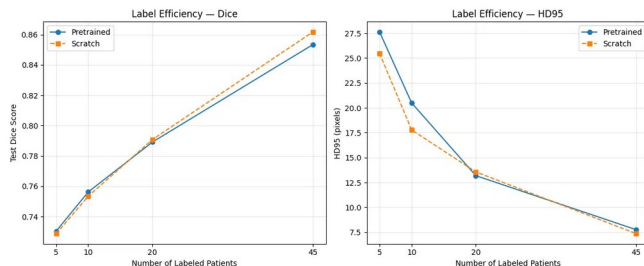


Fig. 5. Label-efficiency curves: test Dice (left) and HD95 (right) as a function of the number of labeled patients.

Training curves for the reduced-label conditions (Fig. 6) reveal a consistent pattern: the pretrained model starts at a much higher validation Dice in early epochs (around 0.5--0.6 at epoch 1 versus 0.1 for scratch at 5 patients), but scratch catches up by approximately epoch 20--30 and reaches comparable or slightly better final performance.

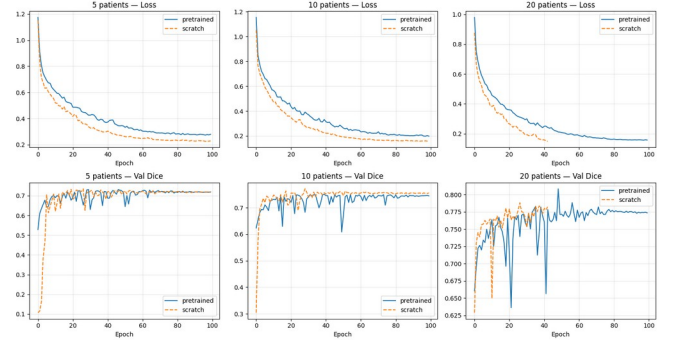


Fig. 6. Training curves at 5, 10, and 20 labeled patients. Top: loss. Bottom: validation Dice. Solid: pretrained. Dashed: scratch.

5. DISCUSSION

5.1. Full-label performance

With all 45 labeled patients available, the scratch U-Net slightly outperforms the pretrained model (Dice 0.862 vs. 0.853). This is not unexpected. When ample labeled data is available, random initialization followed by end-to-end supervised training can learn task-specific features directly, while the pretrained encoder may carry a reconstruction-oriented bias that is slightly suboptimal for segmentation. The small margin between the two models suggests that pretraining neither helps nor hurts significantly in the high-label regime.

5.2. Label efficiency

The hypothesis that pretraining would provide the largest advantage when labels are scarce is only weakly supported by the Dice results and contradicted by the HD95 results. At 5 and 10 patients, the pretrained model has marginally better Dice but worse HD95, meaning it captures slightly more of the prostate area but makes larger boundary errors. One explanation is that reconstruction-based pretraining teaches the encoder about image texture and global structure, which helps with region overlap, but the learned features are not boundary-focused, so the metric most sensitive to boundary errors (HD95) does not benefit.

The training curves tell a more informative story. At 5 patients, pretrained starts at a validation Dice of roughly 0.5 while scratch starts near 0.1. Scratch catches up by epoch 20--30. This suggests that the primary benefit of pretraining is faster convergence rather than improved final performance.

In practical settings with limited compute budgets or early stopping, this faster convergence could still be valuable.

The overall differences across all label conditions are small (less than 1% Dice) and likely within noise for a test set of 20 patients (758 slices). A stronger conclusion would require either a larger dataset or multiple random seeds per condition.

5.3. Autoencoder design choices

A key design decision was omitting skip connections in the autoencoder decoder while including them in the U-Net decoder. With skip connections, the encoder could pass raw pixel information directly to the decoder, bypassing the bottleneck and producing a shallow representation. Without them, the bottleneck at $32 \times 32 \times 512$ must encode all the information needed for reconstruction, encouraging more semantically meaningful features. This mirrors the reasoning in Imran et al. [2], who used a similar convolutional autoencoder architecture for feature extraction from micro-US images.

5.4. High variance in test metrics

Both models show high standard deviations in Dice and HD95, driven by edge slices at the top and bottom of each volume where the prostate occupies a small or negligible fraction of the image. The qualitative examples (Fig. 4) illustrate this: mid-volume slices with a clear prostate region produce strong segmentations from both models, while edge slices with minimal prostate content yield near-empty predictions. A per-patient aggregation (computing Dice per volume rather than per slice) would yield tighter confidence intervals and better reflect clinical performance.

5.5. Future work

The results suggest that a standard reconstruction-based autoencoder may not be the strongest self-supervised objective for segmentation. A fully self-supervised approach such as a masked autoencoder, where random patches are masked and the model must reconstruct only the missing regions, would force the encoder to learn spatial relationships and context rather than simply copying pixels. Transformer-based architectures could also be explored for the pretraining stage, learning richer feature representations through self-attention over image patches, with those features then passed to a segmentation head. Contrastive methods treating different slices of the same patient as positive pairs are another direction. Additionally, the SSL dataset here is limited to 55 patients (2,152 slices), which may not provide enough signal for pretraining to show a clear advantage. Applying this approach to a larger unlabeled pool while keeping the labeled set small would be a fairer test of the hypothesis. The dataset also includes non-expert annotations

from three additional annotators, which could be leveraged for semi-supervised training or uncertainty estimation.

6. REFERENCES

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